Predicting Global Life Expectancy From Air Pollution

Tahsin Moin
Department of Computer Science and Engineering
BRAC University
Dhaka, Bangladesh
tahsin.moin@g.bracu.ac.bd

Maimuna Anjum
Department of Computer Science
Brac University
Dhaka, Bangladesh
maimuna.anjum@g.bracu.ac.bd

Md. Samsul Arafin Anan
Department of Computer Science
Brac University
Dhaka, Bangladesh
samsul.arafin.anan@g.bracu.ac.bd

***Abstract*—Air pollution is a major environmental risk factor that adversely affects human health and longevity, with fine particulate matter (PM2.5) playing a particularly critical role. This paper investigates the relationship between air pollution and global life expectancy using machine learning–based regression models. A country-level dataset was constructed by integrating PM2.5 concentrations with demographic and health-related indicators. After performing data cleaning, feature selection, and preprocessing, a comprehensive set of regression models was implemented and evaluated, including Linear Regression, Ridge Regression, Decision Tree, k-Nearest Neighbors, Random Forest, Extra Trees, Gradient Boosting, AdaBoost, XGBoost, and a neural network–based Multilayer Perceptron (MLP). Model performance was assessed using the coefficient of determination (R^2), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE). Experimental results demonstrate that ensemble and boosting models significantly outperform linear and distance-based approaches, with XGBoost achieving the best performance (R^2 = 0.963, RMSE = 1.79). Analysis of prediction behavior further confirms the strong generalization capability of tree-based ensemble models. The findings highlight the importance of modeling non-linear relationships in environmental health data and demonstrate the effectiveness of advanced ensemble learning techniques for predicting life expectancy under varying air pollution conditions.**

***Index Terms*—Life expectancy, air pollution, PM2.5, machine learning, regression, ensemble models**

I. INTRODUCTION

Life expectancy is a widely used indicator for assessing the overall health status and socio-economic development of populations. It reflects the combined effects of healthcare quality, living conditions, environmental factors, and public health policies. Although global life expectancy has improved over recent decades, environmental pollution remains a significant challenge that continues to adversely affect human health outcomes. Among various pollutants, fine particulate matter with a diameter of 2.5 micrometers or less (PM2.5) is considered one of the most harmful due to its ability to penetrate deep into the respiratory system and bloodstream.

Long-term exposure to PM2.5 has been strongly associated with increased risks of cardiovascular diseases, respiratory illnesses, and premature mortality. Consequently, air pollution has become a critical public health concern, particularly in rapidly urbanizing and industrializing regions. However, accurately quantifying the impact of air pollution on life expectancy is complex, as life expectancy is influenced by multiple interacting demographic, environmental, and health-related factors. Traditional statistical approaches often struggle to capture these non-linear relationships effectively.

Recent advancements in machine learning have enabled more accurate modeling of complex and high-dimensional datasets. Regression based machine learning models, especially ensemble and boosting methods, have demonstrated strong predictive performance in environmental and health - related studies. Motivated by these developments, this study applies and compares multiple machine learning regression models to predict global life expectancy using PM2.5 concentrations and related features. The objective is to evaluate model performance, analyze predictive accuracy, and gain insights into the relationship between air pollution and life expectancy.

II. RELATED WORK

A. Air Pollution and Life Expectancy

The relationship between air pollution and human health has been extensively studied in epidemiology and public health research. Previous studies have consistently reported that long-term exposure to fine particulate matter (PM2.5) is associated with increased risks of cardiovascular diseases, respiratory illnesses, and premature mortality [1] [2]. Several large-scale analyses have demonstrated that elevated PM2.5 concentrations contribute to reductions in average life expectancy at both regional and global levels [3] [4]. These studies establish air pollution as a critical environmental determinant of population health, motivating further investigation using data-driven modeling approaches.

B. Statistical and Regression-Based Approaches

Traditional methods for analyzing the impact of air pollution on life expectancy have primarily relied on statistical regression models [5], including linear and multivariate regression techniques. While these approaches provide interpretability and insights into individual predictors, they often assume linear relationships and are limited in capturing complex interactions among environmental, demographic, and health-related variables. As a result, their predictive performance can be insufficient when applied to heterogeneous, real-world datasets.

C. Machine Learning in Environmental Health Studies

In recent years, machine learning techniques have been increasingly applied to environmental and health-related prediction tasks. Regression-based machine learning models such as decision trees, k-nearest neighbors, and neural networks have shown improved performance over traditional statistical models by capturing non-linear relationships in data [6] [7]. Several studies have demonstrated the effectiveness of machine learning for predicting health indicators, including mortality rates and disease prevalence, using environmental and socio-economic features.

D. Ensemble and Boosting Methods

Ensemble learning methods, particularly tree-based ensembles such as Random Forest and boosting algorithms, have gained popularity due to their robustness and strong generalization capabilities. Random Forest reduces variance through bagging, while boosting methods such as Gradient Boosting and AdaBoost iteratively focus on hard-to-predict samples to reduce bias . More advanced implementations, such as XGBoost, further enhance performance through regularization and optimized training procedures. Prior research has shown that these models often outperform single learners and linear models on structured tabular data, making them well-suited for life expectancy prediction tasks [6] [8].

E. Research Gap and Motivation

Although existing studies have explored the effects of air pollution on health outcomes, there is limited work that systematically compares a wide range of machine learning regression models for predicting global life expectancy using PM2.5 alongside demographic and health indicators. In particular, comparative analyses involving both classical and advanced ensemble models remain scarce. This study addresses this gap by implementing and evaluating multiple regression models within a unified framework to identify the most effective approaches for modeling the relationship between air pollution and life expectancy.

III. Dataset Description

The dataset used in this study was constructed by merging two publicly available datasets related to health and environmental indicators at the country level. The first dataset was obtained from the World Health Organization (WHO) Life Expectancy dataset, available on Kaggle, which contains health, economic, and demographic indicators such as life expectancy, adult mortality, infant deaths, alcohol consumption, BMI, healthcare expenditure, and immunization coverage for multiple countries across several years [9]. The second dataset was sourced from the World Bank Open Data repository and provides country-wise measurements of PM2.5 air pollution (μg/m³), representing fine particulate matter exposure [10]. This environmental indicator was incorporated to analyze the impact of air quality on life expectancy.Both datasets were merged using country name and year as common keys.

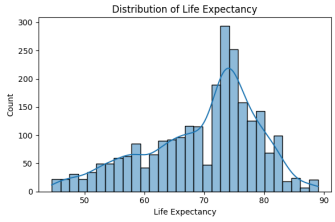


Fig. 1. Distribution of life expectancy

The final dataset contains numerical attributes representing air quality metrics, health-related indicators, and socio-demographic variables, while life expectancy serves as the target variable.The final dataset consists of 2,500 country-year records with 17 features, covering multiple countries across several years. Prior to modeling, the dataset was examined for completeness and consistency. Missing values were handled using appropriate imputation techniques, and all features were converted into a suitable numerical format for machine learning models. Since the dataset consists of variables with different scales and units, feature scaling was applied where required to ensure fair model training, particularly for distance-based and gradient-based algorithms.

The dataset was divided into training and testing subsets to enable unbiased model evaluation. Overall, the dataset provides a comprehensive basis for studying the relationship between air pollution and life expectancy and supports the application of various regression-based machine learning techniques.

IV. Methodology

A. Data Preprocessing

The dataset used in this study was constructed by merging two separate datasets containing air pollution indicators and health-related demographic information. The merged dataset was examined for missing values, inconsistencies, and redundant attributes prior to model development. Records with missing target values were removed, while remaining missing feature values were handled using appropriate imputation techniques to preserve data integrity. The relationship between PM2.5 concentration and life expectancy was further explored using a scatter plot. The visualization indicates a general negative trend, suggesting that higher levels of air pollution are associated with lower life expectancy. The presence of dispersion and non-linear patterns motivated the use of tree-based and ensemble regression models in addition to linear approaches. To analyze interdependencies among key variables, a correlation heatmap was generated for selected health and environmental features. The results show strong correlations between life expectancy and indicators such as adult mortality, schooling, and HIV/AIDS prevalence, while PM2.5 exhibits a moderate negative correlation. These observations informed feature selection and motivated the use of regularized and ensemble models to handle correlated predictors. All features were converted into numerical format to ensure compatibility

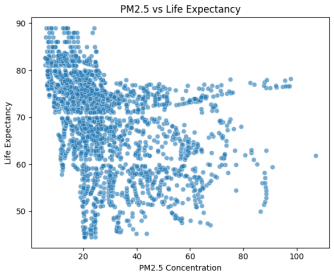


Fig. 2. Relationship between PM2.5 concentration and life expectancy

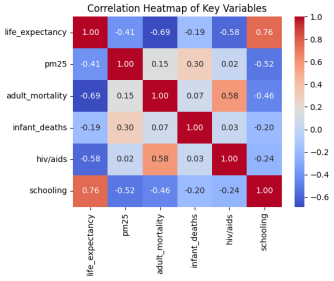


Fig. 3. Correlation heatmap of key variables

with regression-based machine learning models. Since the dataset contains variables measured on different scales, feature scaling was applied where necessary. In particular, standardization was performed for distance-based and gradient-based models such as k-Nearest Neighbors, Ridge Regression, and the Multilayer Perceptron (MLP). Tree-based models were trained on the original feature scales, as they are insensitive to feature magnitude. Finally, the dataset was split into training and testing subsets to enable unbiased evaluation of model performance.

B. Machine Learning Models

To comprehensively analyze the relationship between air pollution and life expectancy, a diverse set of regression models was implemented. Linear Regression was used as a baseline model to capture linear relationships between predictors and the target variable. Ridge Regression was employed to address potential multicollinearity through L2 regularization. A Decision Tree Regressor was included to model non-linear patterns using hierarchical feature splits. In addition, several ensemble and boosting methods were applied to improve predictive performance. Random Forest and Extra Trees regressors were used to reduce variance through ensemble learning, while Gradient Boosting and AdaBoost focused on sequentially minimizing prediction errors. XGBoost, an optimized gradient boosting framework, was implemented to leverage regularization and efficient tree construction. Finally, a neural network–based Multilayer Perceptron (MLP) regressor was trained to represent a deep learning approach and provide a comparison against tree-based models.

C. Model Training and Evaluation Strategy

All models were trained using the same training dataset and evaluated on a held-out test set to ensure fair comparison. Model performance was assessed using standard regression metrics, including the coefficient of determination (R^2), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE). These metrics provide complementary insights into model accuracy, prediction error magnitude, and robustness. Hyperparameters for ensemble and neural network models were selected empirically to balance model complexity and generalization performance. The results obtained from all models were systematically compared to identify the most effective approaches for predicting life expectancy based on air pollution and health-related indicators. A single fixed train-test split was used for all models to ensure fair and consistent comparison.

D. Tools and Implementation

All experiments were implemented in Python using the Google Colab environment. Core libraries included pandas and NumPy for data manipulation, scikit-learn for machine learning model implementation and evaluation, and XGBoost for advanced gradient boosting. Visualization libraries such as Matplotlib and Seaborn were used for exploratory data analysis and result presentation. This toolchain enabled efficient experimentation, reproducibility, and clear performance comparison across models.

V. Results & Evaluation

This section presents the experimental results obtained from the implemented machine learning models and provides a comparative evaluation of their predictive performance. All models were evaluated on the same held-out test dataset to ensure a fair comparison. Standard regression metrics, including the coefficient of determination (R^2), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE), were used to assess model accuracy, error magnitude, and robustness.

A. Model Performance Comparison

To compare the overall predictive capability of the models, a summary table of performance metrics was generated. The results indicate a clear performance gap between linear models and advanced ensemble methods. Linear Regression and Ridge Regression achieved relatively lower R^2 scores (0.82), suggesting limited ability to capture complex relationships in the data. In contrast, tree-based and ensemble models demonstrated substantially stronger performance. Among all evaluated models, XGBoost achieved the highest predictive accuracy, with an R^2 score of 0.9629, RMSE of 1.79, and MAE of 1.07, indicating excellent generalization. Extra Trees and Random Forest also performed competitively, achieving R^2 scores above 0.95. Gradient Boosting exhibited comparable performance, confirming the effectiveness of boosting strategies for this task. The k-Nearest Neighbors model exhibited signs of overfitting, achieving perfect training performance but

	Model	R2	RMSE	MAE
8	XGBoost	0.962900	1.789938	1.071920
5	Extra Trees	0.962085	1.809487	1.131253
4	Random Forest	0.957792	1.909186	1.196094
6	Gradient Boosting	0.957284	1.920644	1.257284
2	Decision Tree	0.913829	2.727907	1.951212
3	kNN	0.892711	3.043878	2.047470
9	MLP Regressor	0.886315	3.133299	2.004301
7	AdaBoost	0.876420	3.266808	2.541377
1	Ridge Regression	0.815547	3.991097	2.986767
0	Linear Regression	0.815542	3.991156	2.986802

Fig. 4. Model performance comparison on the test set

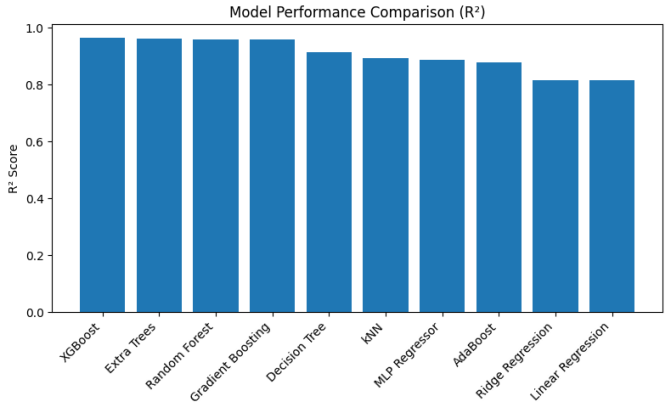


Fig. 5. Model performance comparison using R^2 score

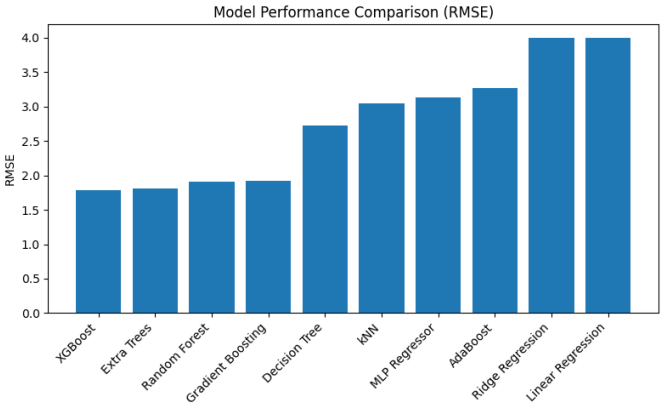


Fig. 6. Model performance comparison using RMSE

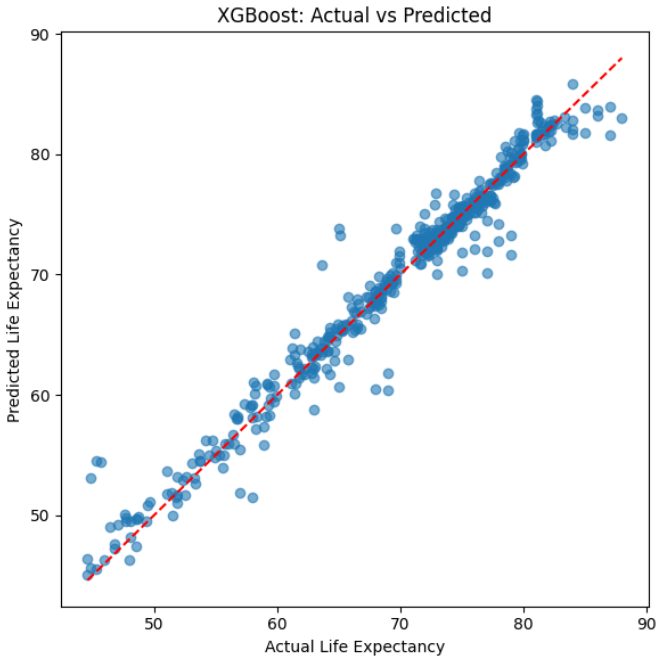


Fig. 7. XGBoost actual vs. predicted life expectancy

reduced accuracy on the test set, which is consistent with its instance-based learning nature. Neural network models, such as the Multilayer Perceptron (MLP), showed moderate performance, while AdaBoost produced comparatively higher prediction errors.

B. Performance Metrics Visualization

To visually compare model performance, bar plots of R^2 and RMSE scores were generated. Fig. 5 illustrates the R^2 scores for all models, clearly highlighting the superiority of ensemble and boosting methods over linear approaches. XGBoost and Extra Trees achieve the highest R^2 values, confirming their strong predictive capability. Fig. 6 presents the RMSE comparison across models. Lower RMSE values indicate better predictive accuracy, and the results are consistent with the R^2 analysis. XGBoost records the lowest RMSE, followed closely by Extra Trees and Random Forest, while linear models exhibit significantly higher error values.

C. Actual vs. Predicted Analysis

To further analyze model behavior, an actual versus predicted scatter plot was generated for the best-performing model, XGBoost. As shown in Fig. 7, the predicted life expectancy values closely align with the ideal diagonal line, indicating strong agreement between actual and predicted values. The tight clustering of points around the reference line demonstrates low prediction error and strong generalization capability across different life expectancy ranges. Minor deviations are observed at extreme values, which is expected in real-world data.

D. Overall Analysis

Overall, the results demonstrate that tree-based ensemble and boosting models consistently outperform linear,

distance-based, and neural network approaches for predicting life expectancy. The strong performance of XGBoost and Extra Trees highlights the importance of modeling non-linear relationships and feature interactions in environmental health data. Since life expectancy is a continuous variable, regression metrics such as R^2, RMSE, and MAE were used instead of classification metrics like accuracy or precision. These findings confirm that advanced ensemble learning techniques are well-suited for capturing the complex interplay between air pollution, health indicators, and life expectancy.

VI. Discussion

The experimental results demonstrate that machine learning models are effective in predicting life expectancy using air pollution, health, and socio-economic indicators. In particular, tree-based ensemble and boosting methods consistently outperformed linear, distance-based, and neural network models. The superior performance of XGBoost and Extra Trees highlights their ability to capture complex non-linear relationships and feature interactions present in environmental health data. The strong alignment between actual and predicted values for XGBoost further confirms its robustness and generalization capability.Despite these promising results, several limitations should be acknowledged. First, the dataset was constructed by merging two independent sources, which may introduce inconsistencies or alignment errors despite careful preprocessing. Second, the analysis relies on country-level aggregated data, which may mask regional or individual-level variations in pollution exposure and health outcomes. Additionally, hyperparameter tuning was performed empirically rather than through exhaustive optimization, which may limit the maximum achievable performance of some models. Moreover,These findings may support policymakers in identifying high-risk regions and prioritizing interventions aimed at reducing air pollution and improving public health outcomes. Future improvements could include incorporating additional environmental variables, such as ozone levels and applying advanced hyperparameter optimization techniques. Exploring temporal models or region-specific analyses may also provide deeper insights into the dynamic relationship between air pollution and life expectancy.

VII. Conclusion

This study investigated the effectiveness of various machine learning regression models for predicting life expectancy using air pollution, health, and socio-economic indicators. A comprehensive comparison of linear, tree-based, ensemble, boosting, and neural network models was conducted using standard regression metrics. The experimental results demonstrate that ensemble and boosting methods, particularly XGBoost and Extra Trees, significantly outperform traditional linear and distance-based models. The strong predictive performance of these models highlights their ability to capture complex non-linear relationships and feature interactions present in environmental health data. Despite achieving high predictive accuracy, the study is subject to certain limitations, including reliance on country-level aggregated data and a limited set of environmental variables. Future work may focus on incorporating additional air quality indicators, performing region-specific or temporal analyses, and applying advanced hyperparameter optimization techniques. Integrating deep learning models with richer spatial and temporal data could further enhance predictive performance and provide deeper insights into the impact of air pollution on life expectancy.

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